Large-Area Land-Cover Changes Monitoring With Time-Series Remote Sensing Images Using Transferable Deep Models

Jining Yan, *Member, IEEE*, Lizhe Wang, *Fellow, IEEE*, Haixu He, Dong Liang, Weijing Song, and Wei Han

***Abstract*— Dense time-series remote sensing images have transformed the traditional bitemporal land-cover change detection to continuous monitoring. Previous work mostly employs linear fitting, prediction, or decomposition methods, and the detection accuracy is not high. The latest progress of deep learning (DL) shows its advantages in time-series change monitoring. However, DL models are computationally expensive and require lots of labeled samples, resulting in often employed prediction-threshold-based unsupervised change detection method. However, the determination of a reasonable threshold has always been a big problem. Therefore, we proposed the similarity-measurement-based deep transfer learning for time-series adaptive change detection (SDTL-TSACD) model. First, a standard dynamic time warping (SDTW) distance was proposed and used to cluster large-scale time series into multiple subcategories with high time-series similarity. Second, a time convolutional network (TCN) was used for nonlinear time-series fitting and prediction, and an early stop strategy was used to prevent overfitting. Then, the trained TCN model would be transferred and performed pixel-by-pixel time-series prediction within the same category, and the SDTW was also used to evaluate the prediction accuracy. Finally, the Otsu adaptive threshold was used to detect change points, and the spatial neighbor relationship was used to eliminate the pseudo-change points. Change detection results using 132 benchmark datasets showed that the SDTL-TSACD performed well in both accuracy and efficiency. In addition, the MOD13Q1-EVI images from 2001 to 2020 were used to study the land-cover change of the Loess Plateau, and the SDTL-TSACD also showed a good ability to solve practical problems.**

***Index Terms*— Large-area land cover, standard dynamic time warping (SDTW), time convolutional network (TCN)-Otsu, time-series changes monitoring, transferable deep models.**

Manuscript received October 26, 2021; revised January 23, 2022 and March 4, 2022; accepted March 15, 2022. Date of publication March 17, 2022; date of current version April 12, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41925007 and Grant U21A2013 and in part by the Open Research Project of the Hubei Key Laboratory of Intelligent Geo-Information Processing under Grant KLIGIP-2021B10. *(Corresponding author: Lizhe Wang.)*

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Digital Object Identifier 10.1109/TGRS.2022.3160617

I. Introduction

With the accumulation of historical Earth observation (EO) data, large-scale and long-term remote sensing data have played an increasingly important role in global climate change [1], forest disturbance monitoring [2], global urban expansion tracking [3], and land-use/land-cover change detection [4]. Traditional bitemporal land-cover change detection has been transformed to continuous time-series remote sensing monitoring. Dense time-series analysis has also become a research hotspot in remote sensing technology and applications in recent years [5]. In the current era of rapid social and economic development, it is critical to use continuous remote sensing data to rapidly determine the precise time at which the land-cover changes in a large area, clarify the drivers of this change, and provide managers with timely and effective change information and decision-making suggestions.

However, large-scale time-series remote sensing monitoring still has the following problems.

1) The large-scale [6], nonlinear [7], nonstationary, and other big data characteristics of dense time-series remote sensing images make it difficult to capture complex and multidirectional land-cover transitions [8]. The traditional techniques often assume that time-series changes are a smooth process and mostly employ linear fitting, prediction, or decomposition methods, which may identify many pseudo-change points [9]. Therefore, large-scale remote sensing time-series change detection mostly adopts the sampling analysis mode, and there are few successful cases of full data analysis using dense time-series remote sensing image [10]. However, long-term, continuous, and uninterrupted EOs can capture many fine features of land-cover changes, and the sampling analysis will inevitably fail to fully identify frequent or short-term land-cover changes caused by natural phenomena and human activities.
2) The deep learning (DL) model does not have the assumption of temporal stability and can automatically capture the nonlinear and nonstationary change characteristics, as well as the high-dimensional space–time spectral correlations of time-series remote sensing images. Therefore, the DL model can completely extract land-cover changes from the time-series remote sensing images [11], [12]. However, the DL model often needs to

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be operated on expensive dedicated graphics processing unit (GPU) servers [13], which has high computational time complexity. In addition, the DL model requires numerous labeled samples, and the domain shift between training samples and test samples can easily cause a significant performance drop [14].

3) The prediction-threshold-based unsupervised change detection method [15] is a strategy frequently adopted in recent years to overcome the problem that DL models require a large number of labeled samples. However, the threshold directly determines the accuracy of change point detection, and a reasonable threshold setting has always been a difficult problem.
4) In addition, the similarity-measurement-based transfer learning may also enable DL to solve the problem of large-scale land-cover change detection [16]. The dynamic time warping DTW) with better noise immunity provides better temporal similarity measure. However, the value range of DTW is very wide [17]. If the two time series are very similar, the DTW value is small; if the similarity is very low, the DTW value is very large, making it difficult to determine a reasonable threshold to judge which two time series are similar in large-scale time-series data.

Therefore, this article proposes a transferred DL-based time-series remote sensing monitoring and analysis model, named similarity-measurement-based [18] deep transfer learning for time-series adaptive change detection (SDTL-TSACD), for monitoring large-area land-cover changes, to improve the quality and accuracy of the information derived from remote sensing time-series images.

This model includes the following major contents.

1) Considering the difficulty of time-series modeling caused by the complex and multidirectional land-cover transitions, a time convolutional network (TCN) is used for nonlinear time-series fitting and prediction, and an early stop strategy is used to prevent overfitting.
2) In order to solve the problem that the transferable range of the DL model is difficult to determine, a standard dynamic time warping (SDTW) distance is proposed and used to cluster large-scale time series into multiple small classes with high time-series similarity. The trained TCN model can be transferred and performed pixel-by-pixel time-series prediction within the same category, and the SDTW is also used to evaluate the prediction accuracy. If the prediction accuracy of the transfer DL model is less than the preset SDTW threshold, it can be directly used for change point detection; otherwise, it needs to be divided into new subcategories to repeat the model training and evaluation process.
3) For the problem that a reasonable threshold of change point is difficult to determine, the Otsu adaptive threshold is used to judge the difference between the predicted and observed values to detect the time-series change points. In addition, the pseudo-change points are eliminated using the spatial neighbor relationship, which not only ensures that the change points will not be missed but also improves the detection accuracy.

Finally, 132 benchmark datasets are used to verify the accuracy of change point detection, and the dense time-series MODIS MOD13Q1 enhanced vegetation index (EVI) data from 2001 to 2020, collected in Loess Plateau, are used to verify the practicality of the SDTL-TSACD model in solving the problem of large-area land-cover change detection.

II. Related Work

Time-series remote sensing monitoring focuses on the land-cover changes in the same area captured by long-term continuous observations, which often uses methods such as fitting, forecasting, difference, and similarity analysis of historical EO data for change point detection. In summary, remote sensing time-series change detection can be roughly divided into postclassification, direct classification, image transformation-based, similarity threshold-based, model-based, and DL-based change detection methods.

A. Postclassification Change Detection Method

The postclassification change detection method aims to combine, segment, and classify the remote sensing images obtained at a certain time point or time interval to obtain the land-cover classification results and then compare them with the classification results at other time points or time intervals to detect change points [19]. Gong [20] used the object change detection algorithm to extract and classify land-cover features from the SPOT-VEGETATION time series and performed long-term change detection and analysis on the yearly classified results. Linke and McDermid [21] used the object classification algorithm to classify the land cover from time to time and then used the classification results to perform multitime comparison and change detection. Wang *et al.* [22] used the band synthesis method to synthesize 23 bands of MODIS normalized difference vegetation index (NDVI) within a year into a 23-band image by using algorithms such as unsupervised classification and decision tree to generate land-cover classification results for each year. Then, they performed postclassification change detection. Although this method is simple, the workload of classifying the ground features by the time phase is considerable. Moreover, the classification process does not consider the dependent characteristics of various ground features in the time dimension, and its comparison with time-phase classification can cause error accumulation [23].

B. Direct Classification Change Detection Method

The direct classification change detection method aims to directly input the time-series remote sensing data or their statistical and spectral characteristics into the classifier to perform binary classifications of changed and unchanged data. Woodcock *et al.* [24] used an artificial neural network method to classify Oregon's forest remote sensing data into changed and unchanged data and realized the detection of forest-cover time-series changes. Wang *et al.* [25] used segmentation and classification methods to extract high-precision change information from high-resolution airborne and SPOT5 remote sensing images. Gil-Yepes *et al.* [26] used the decision tree classification algorithm to input the semivariance, cross semivariance, and other statistical and spectral features of the

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cropped time-series remote sensing data of Murcia in southeastern Spain into the classifier for direct classification change detection. This method avoids the accumulation of errors in the postclassification change detection method. However, it requires numerous accurately changed and unchanged samples, which is not suitable for large-scale time-series remote sensing change detection applications.

C. Image Transformation-Based Change Detection Method

In the image transformation-based change detection method, by performing principal component transformation and tasseled cap transformation on the original time-series remote sensing data or its statistical and spectral characteristics, the change information is enhanced and identified. Gong [20] extracted the change information by performing a principal component transformation on the differential time-series remote sensing data. Deng *et al.* [27] used principal component transformation to extract the change characteristics of time-series remote sensing data and comprehensively used multisource remote sensing data and their hybrid classification method to obtain high-precision land-cover change information. Allen *et al.* [28] used tasseled cap transformation and change vector analysis methods to achieve 25-year vegetation change information enhancement in the Montado Forest Ranch in southern Portugal and then effectively identified the characteristics of the temporal changes. However, the image transformation-based change detection method does not completely consider the physical meaning of each transformed component and the spatial distribution characteristics of the ground features, and it is often impossible to obtain accurate land-cover change detection results.

D. Similarity Threshold-Based Change Detection Method

In the similarity threshold-based change detection method, a reference time series is set, and the similarity of other time series with the reference time series is sequentially compared. If the similarity is lower than a certain threshold, a change is considered to have occurred. This method can be roughly classified into two types. The first type sets the first cycle of the original time-series remote sensing data as the reference time series and then compares the similarity between the subsequent cycle and the first cycle. If the similarity is lower than the preset threshold, it is considered that land-cover has occurred. The second type uses data fitting and prediction models to predict the time-series remote sensing data and then compares the similarity of the difference between the predicted and actual values. If the difference is higher than the preset threshold, a change is considered to have occurred. The similarity measurement generally uses Euclidean distance, Jeffries–Matusita (J–M) distance, and DTW distance. For example, Radoi and Burileanu [29] used DTW distance and expectation-maximization-based thresholding to evaluate the similarity of satellite image time series. Vorovencii *et al.* [30] analyzed the land-cover changes in Cibiau County, Romania, by measuring the difference and similarity of NDVI and other vegetation indexes using the Euclidean distance. Huang *et al.* [31] calculated the similarity of the annual synthetic NDVI time series in Beijing from 1985 to 2015 and set thresholds to obtain the spatial distribution of vegetation increase and decrease. Gong *et al.* [32] mapped a global impervious surface using a long time sequence NDVI, modified normalized difference water index (MNDWI), and shortwave infrared bands with the help of a thresholding method. The similarity threshold-based change detection method is simple and can be directly applied to various remote sensing applications; however, it requires users to distinguish and choose among multiple similarity measures according to the characteristics of the problem.

E. Model-Based Change Detection Method

The model-based change detection method takes statistical assumptions as the premise and comprehensively uses regression, fitting, prediction, and other methods to construct a spatiotemporal evolution model to realize time-series remote sensing data change detection. Typical time-series remote sensing data change detection models include the breaks for additive season and trend algorithm [33], Landsat-based detection of trends in disturbance and recovery algorithm [34], continuous land-cover change detection and classification model [35], detecting breakpoints and estimating segments in trend model [36], subannual change detection model [37], continuous monitoring of land disturbance model [38], and prophet [23].

This method can effectively overcome the impact of factors such as terrain differences, illumination, and noise on time-series remote sensing data and obtain relatively accurate change point information. However, the model construction is relatively difficult, and each model has its own applicable data source and data characteristics, which is not suitable for the direct change detection of large-scale time-series remote sensing data.

F. DL-Based Change Detection Method

The DL-based change detection method aims to completely exploit the end-to-end automatic learning mechanism of the DL model and realize high-precision timing change detection without relying on considerable prior knowledge. Lyu *et al.* [39] used a recurrent neural network (RNN) to study multitemporal land-use change detection and verified the superiority of the method through binary classification, migration, and multiclass transformation experiments. Zhong *et al.* [40] comprehensively used long- and short-term memory (LSTM) networks and convolutional neural networks (CNNs) to obtain better time-series classification effects than support vector machines (SVMs) and random forests. Zhang *et al.* [41] used multilayer perceptrons and CNNs to mine the spectral and spatial characteristics of time-series remote sensing data and then obtained more accurate land-cover change detection information. Ienco *et al.* [42] used a multisource DL framework to comprehensively consider the space and time dependence of time-series remote sensing data and used this information for the land-cover change detection research of Sentinel-1 and 2 optical remote sensing data. Yan *et al.* [43] used the TCN model to conduct multitemporal research on land-cover prediction in Wuhan and verified the superiority of this model by comparing the experimental results with those obtained using traditional models.

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DL models require numerous training samples and computing resources, and the deep network structure and transferability performance need to be investigated.

Through the above review of the commonly used time-series remote sensing monitoring methods, the following conclusions can be drawn.

1) The traditional remote sensing time-series change detection methods often assume that the time series is a stationary process. The detection accuracy of nonlinear and nonstationary time-series changes caused by frequent land-cover evolution is not high, and many pseudo-change points can be easily detected.
2) The large-scale characteristics of time-series remote sensing data are bound to pose huge challenges for training samples and computing resources for DL models. However, the types of ground cover are complex, and it is difficult to artificially generate training samples with various ground features. In addition, the computational cost of the DL model is extremely high, and the pixel-by-pixel model training will inevitably consume considerable computing resources.

Therefore, using the similarity characteristics of remote sensing time series, performing DL model migration in a time series with high similarity to reduce the number of training to the maximum possible extent, and calculating the similarity between the predicted results of the transferred model and observation data to ensure the accuracy of the migration are inevitable methods to improve the efficiency of the DL model for large-scale time-series remote sensing monitoring.

III. Methods

The transferred DL-based change detection model (i.e., SDTL-TSACD) mainly includes: 1) TCN-based remote sensing time-series prediction; 2) similarity-measurement-based DL model migration; and 3) Otsu-based adaptive time-series change detection. The overall technical solution for the SDTL-TSACD is shown in Fig. 1.

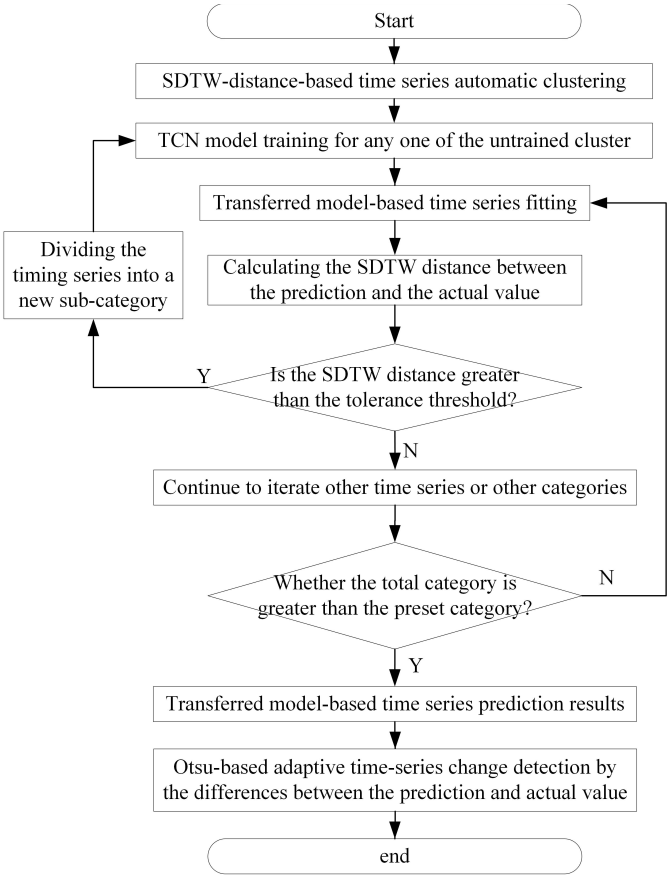

Fig. 1. Overall technical solution of the SDTL-TSACD.

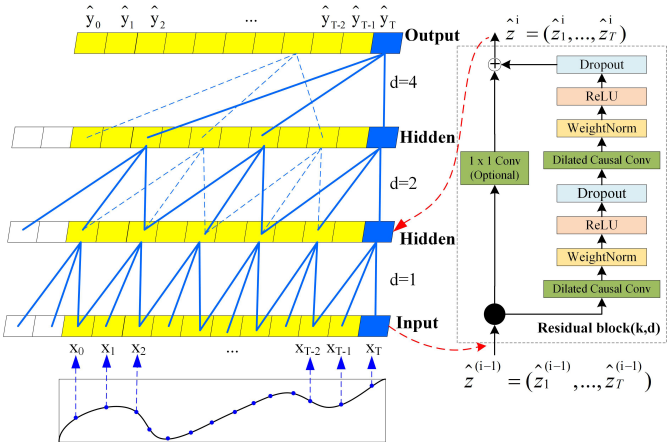

Fig. 2. TCN, including a dilated causal convolution with dilation factors $d = 1, 2$, and 4 and a filter size $k = 3$, and residual block. In the dilated causal convolution part, $x_0, \ldots, x_T$ are the input sequence, and $y_0, \ldots, y_T$ are the corresponding output. In the residual block, $\widehat{z}_t^i$ represents the value corresponding to the ith hidden layer at time t.

A. TCN-Based Remote Sensing Time-Series Prediction

After denoising processing, the remote sensing time series becomes closer to the real growth conditions of the ground objects as a whole. The methods, such as statistical models, machine learning, and DL, can be directly used to fit the change trend, to compare the difference between the predicted and actual values, for change point detection. We intend to use TCN to achieve high-precision prediction of time-series remote sensing data. As a DL model that specializes in dealing with long-term timing dependencies, TCN performs better than timing models, such as RNN and LSTM, due to its simple model and high efficiency. TCN exhibits the following features: 1) the length of the input sequence is arbitrary and the lengths of the output and input sequences are consistent and 2) the convolution layer adopts causal convolution and no information is leaked from the past to the future [44]. To achieve the first feature, TCN uses a 1-D fully convolutional network (FCN) architecture, where each hidden layer is the same length as the input layer, and zero padding of length is added to keep subsequent layers the same length as previous ones [45], [46]. To achieve the second feature (i.e., causal convolution), the output at time t is only convolved with elements in the upper layer at the current time or earlier (Fig. 2).

B. Similarity-Measurement-Based DL Model Migration

For DL-based pixel-by-pixel time-series prediction, a model needs to be trained and predicted separately for each pixel.

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The time complexity of the algorithm, as well as the demand for storage space and computing resources, is high. However, the time-series change rules of the same type of features are often relatively similar, and time-series prediction models with similar change rules can be migrated and used. Therefore, to avoid the time-consuming and labor-intensive problem of pixel-by-pixel DL model training caused by large-scale time-series remote sensing images, this article intends to implement the transferred DL models based on the time-series similarity measurement. First, we propose to use the SDTW distance to compare the similarity of time series and determine the transferable range of the DL model. Thereby, to complete the prediction, the trained DL-based time-series model can be migrated to the untrained model with higher similarity. Then, the SDTW distance is also used to measure the similarity between the predicted result and the actual data. If the tolerable similarity threshold is exceeded, the transferred DL model will be discarded and a new DL model will be trained to obtain better time-series prediction results.

The proposed background and principle of SDTW are given as follows. The DTW distance can dynamically adjust the difference between the curve amplitude and displacement in the nonuniform time series and more accurately measure the similarity of the two time-series curves. However, the classic DTW distance uses the Euclidean distance as the measurement benchmark. The similarity threshold varies considerably for large-scale time-series remote sensing images [17], and it is not convenient to preset the evaluation threshold. This article proposes the SDTW distance and uses the standard Euclidean distance as the benchmark for time-series similarity measurement.

The principle of SDTW is given as follows. Assume that the predicted and actual values of the two time-series vectors are, respectively, $Q = q_1, q_2, \ldots, q_m$ and $C = c_1, c_2, \ldots, c_n$, where m and n represent the lengths of the two vectors. Calculating the similarity between the two vectors is to calculate the regularization function corresponding to the minimum cumulative distance between the characteristic points of the two curves (Fig. 3). In this study, the distance between point i on the Q curve and point j on the C curve is measured using the weighted Euclidean distance; thus,

$$d(q_i, c_j) = \sqrt{(q_i - c_j)^2} \Big/ \text{std}(Q, C) \tag{1}$$

where $\text{std}(Q, C)$ denotes for the standard deviation of vectors Q and C. Then, the cumulative distance can be expressed as

$$\gamma(i, j) = d(q_i, c_j) + \min\{\gamma(i-1, j-1), \gamma(i-1, j), \gamma(i, j-1)\}. \tag{2}$$

The final SDTW distance can be expressed as $\text{SDTW}(Q, C) = \gamma(m, n)$. If $\text{SDTW}(Q, C)$ exceeds the preset threshold, the prediction time sequence is expected to considerably deviate from the observation time sequence, and the migration DL model will be deprecated and a new DL model training needs to be performed.

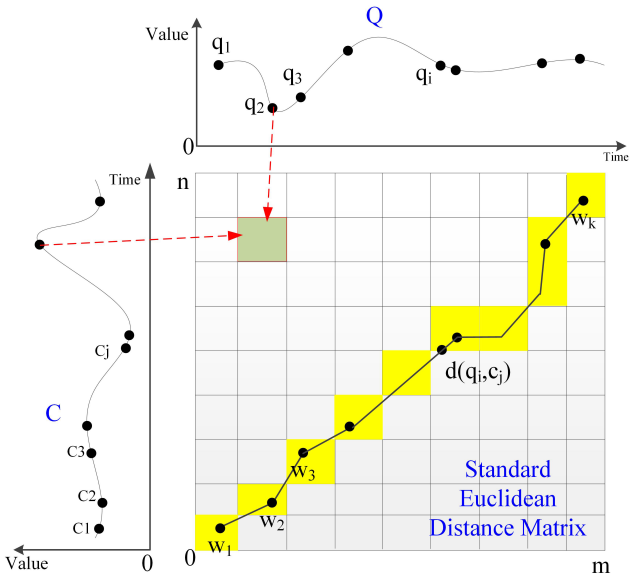

Fig. 3. SDTW-based prediction evaluation of the transferred DL model.

C. Otsu-Based Adaptive Time-Series Change Detection

The change point of the time series (i.e., the average value and mean square error of the entire time series from this point) begins to change, which in turn affects the smooth characteristics of the entire time series. For time-series forecasting, through the learning and regression analysis of historical change trends, future time points are forecast. However, for the change point in the time series, the prediction model is too late to capture the change, and there is often a huge difference between the predicted and actual values at the change point. We can set a change point detection threshold, and if the difference range between the predicted and actual values exceeds the preset threshold, the sequence point is judged as a change point. Therefore, this study intends to use the 1-D adaptive threshold Otsu [47] method to detect changes in large-scale time-series remote sensing images and eliminate pseudo-change points based on the spatial neighbor relationship to prevent false detection and missed detection, to achieve high-precision time-series change detection.

1) Otsu-Based Change Detection: The difference between the predicted and actual values indicates the change points of the time-series remote sensing data. This study intends to learn from the adaptive threshold Otsu method in the field of digital images and improves it to make it suitable for time-series change point detection.

According to the gray-scale characteristics of the difference sequence, the entire sequence is divided into two parts: the background and the change point. As variance is a measure of the uniformity of grayscale distribution, the greater the interclass variance between the change point and the background, the greater is the difference between the two. Once a part of the change point is incorrectly classified as the background or vice versa, the difference between the two parts decreases. Therefore, maximizing the variance between the change points

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and the background value indicates that the probability of misclassification is minimized.

For the absolute difference, $S(t) = s_1, s_2, \ldots, s_t, \ldots, s_N$ between the predicted and observed values, where t is the time and N represents the total length of the sequence. i represents the gray level of the sequence point s_t, n_i represents the number of sequence points with the gray level i, and p_i represents the occurrence probability of a sequence point with gray level i (see the following equation):

$$p_i = \frac{n_i}{N}, \quad i \in [\min\{S(t)\}, \max\{S(t)\}]. \tag{3}$$

Assume that the segmentation threshold between the change point and the background is recorded as τ, and the previous gray level of τ is recorded as $\tau - \varepsilon$, where ε is the histogram grayscale interval of the absolute difference $S(t)$. The sequence point value belonging to the background is $[\min\{S(t)\}, \tau - \varepsilon]$, the proportion of points belonging to the background to the entire time series is recorded as δ_0, and the average gray level is μ_0; the sequence point value belonging to the background is $[\tau, \max\{S(t)\}]$, the proportion of points belonging to the background to the entire time series is recorded as δ_1, and the average gray level is μ_1 (see the following equation):

$$\begin{aligned}
\delta_0 &= \sum_{i=\min\{S(t)\}}^{\tau-\varepsilon} p_i \\
\delta_1 &= \sum_{i=\tau}^{\max\{S(t)\}} p_i = 1 - \delta_0 \\
\mu_0 &= \sum_{i=\min\{S(t)\}}^{\tau-\varepsilon} i \cdot \frac{p_i}{\delta_0} \\
\mu_1 &= \sum_{i=\tau}^{\max\{S(t)\}} i \cdot \frac{p_i}{\delta_1}.
\end{aligned} \tag{4}$$

We establish the objective function $g = \delta_0 \cdot {\mu_0}^2 + \delta_1 \cdot {\mu_1}^2$, where g indicates the variance between the change point and background. When g takes the global maximum, the corresponding T is the optimal threshold. In the absolute difference series between the predicted and observed values, the grayscale histogram shows a unimodal state, and the number of change points is very small, but the value is greater than the average value of the series. Therefore, the contribution of the first item in the objective function g should be appropriately increased and the second item should be decreased. In addition, when the background pixel ratio δ_0 is greater than 0.5, the contribution value of the first item will dominate, resulting in a particularly large segmentation threshold T. Last but not least, it is necessary to introduce an overall weight coefficient to make the segmentation threshold point at the trough position of the gray histogram. In the actual calculation process, in order to make each weight coefficient self-adaptive, we use parameters δ_0 and δ_1 as weighting factors for each item. Therefore, the improved objective function g is shown in the following equation:

$$g = \begin{cases} \omega \cdot \left(\delta_1 \cdot \delta_0 \cdot {\mu_0}^2 + \delta_1 \cdot {\mu_1}^2\right), & \delta_0 \le 0.5 \\ \omega \cdot \left(\delta_0 \cdot \delta_0 \cdot {\mu_0}^2 + \delta_1 \cdot \delta_1 \cdot {\mu_1}^2\right), & \delta_0 > 0.5. \end{cases} \tag{5}$$

In addition, it can be seen from (3) that the proportion p_T of points belonging to the optimal segmentation threshold T will be very small, and a larger overall weight coefficient ω is required. Thus, it would be a good choice to set $\omega = 1 - p_T$. In addition, in order to prevent discontinuity of sequence points, the value of p_T can be replaced by the average value $\overline{p_T}$ of the proportion of sequence points near T (see the following equation):

$$\omega = 1 - \overline{p_T} = 1 - \frac{1}{3} \cdot \sum_{i=T-\varepsilon}^{T+\varepsilon} p_i. \tag{6}$$

2) Spatial Neighbor Relationship-Based Pseudo-Change-Point Elimination: As the judgment of the adaptive threshold change point is entirely based on the temporal difference information, the spatial neighbor relationship of remote sensing data is not considered. Therefore, it is also necessary to use the Hilbert [48] space-filling curve to extract the 2-D plane data composed of each time-phase change point into a 1-D curve and set the linear filtering rule: if there are changes in three or more consecutive pixels, it is considered to be the change point; otherwise, it is considered as a pseudo-change point and should be eliminated to improve the accuracy of timing change detection.

IV. Validation

For dense time-series remote sensing monitoring, the most important thing is to detect all change points and determine the time at which the change points occur. Therefore, to verify the effectiveness of the proposed TCN-Otsu change detection method, 132 MOD13A2 16-day synthetic EVI time series were selected with Google Earth Engine. These benchmark datasets all contain at least one change point, and the time at which the change point occurs is determined by the National Cryosphere Desert Data Center website. These benchmark datasets have been shared to the GitHub website "https://github.com/CUG-BEODL/CUG-FFireMCD" [58]. To ensure the reliability of the prediction results, this study repeats the model training and prediction 100 times independently by using precision, recall, and $F1$-score to measure the average accuracy of change point detection. Precision measures the probability that a change point is correctly identified, and recall rate measures the proportion of correctly identified change points to all change points. The higher the precision and recall rate, the better the model. $F1$-score can be regarded as a weighted average of model precision and recall, with the range of 0–1, which can reflect the overall situation of the model [49].

Fig. 4 presents an ideal change point detection results of the six time series we arbitrarily selected. It should be noted that we assume that a time point before and after the change point is the tolerance interval, and the detected change points located in the tolerance interval will be positively evaluated.

To further illustrate the superiority of the proposed "TCN-Otsu" algorithm, another commonly used DL time-series model LSTM and the commonly used change point detection method "3σ-rule" are selected for comparison and verification. LSTM is a time-RNN suitable for processing and predicting important events with relatively long intervals

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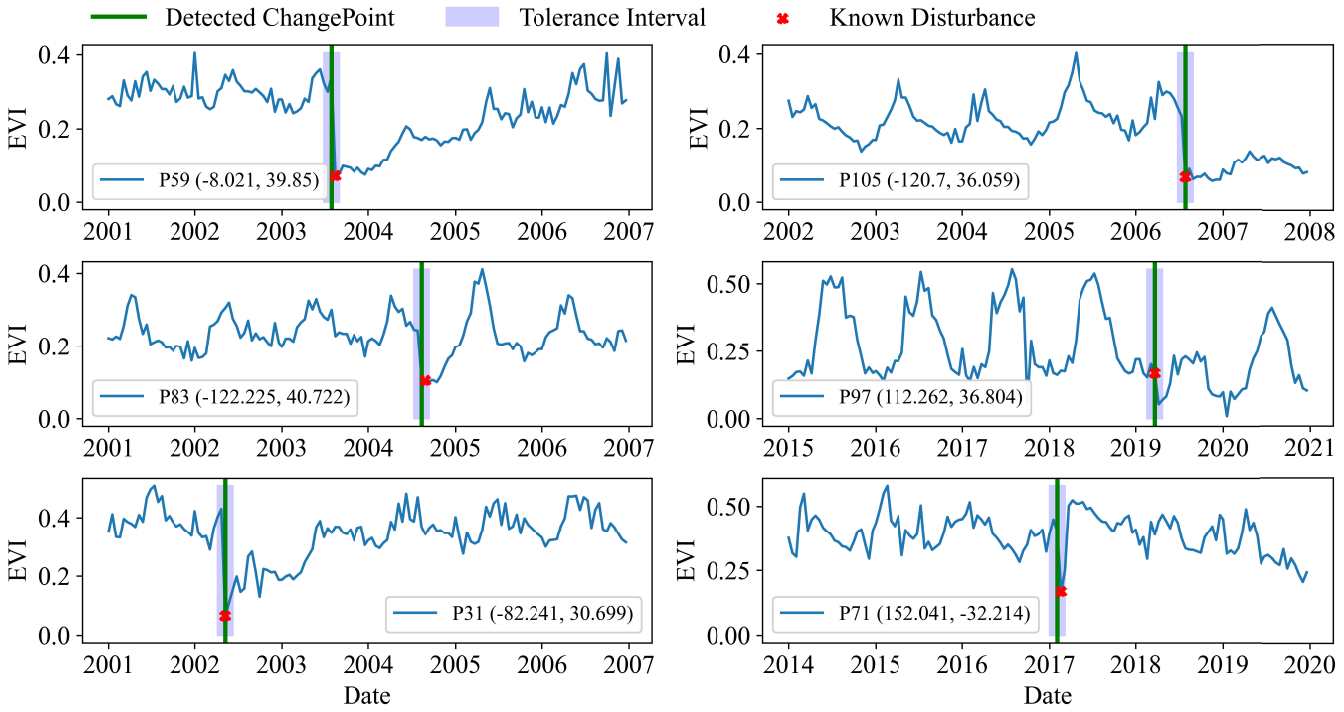

Fig. 4. TCN-Otsu-based change point detection results of time-series benchmark datasets. *P* represents the number of the sampling point, and the values in parentheses represent the latitude and longitude of the point.

and delays in time series. However, its model is complicated and not suitable for handling long time intervals [43]. The "3σ-rule," where σ represents the standard deviation of the difference between the predicted and actual values, that is, the probability that the difference between the predicted and actual values is within the interval of "mean value $\pm$ 3 standard deviations," is 99.74%. When this interval is exceeded, a change point is considered to have occurred [35]. It should be added that in order to prevent the overfitting phenomenon of TCN and LSTM, an early stop strategy [50] was adopt to set the number of model training iterations. The final comparative experiment results are shown in Table I, which shows that the proposed "TCN-Otsu" model outperforms other methods on all experimental evaluation metrics on the benchmark datasets. Specifically, the Otsu adaptive threshold method is more accurate than the "3σ-rule" in change point detection but runs slightly longer; the convergence speed of the TCN model is much faster than that of LSTM, about 45% time saving. Conclusion is that the "TCN-Otsu" model is more suitable for large-scale and long-time-series remote sensing monitoring.

TABLE I

COMPARATIVE EXPERIMENTAL RESULTS OF DIFFERENT CHANGE DETECTION MODELS WITH THE BENCHMARK TIME SERIES

Method	Average precision	Average recall	Average F1-score	Average time(s)
TCN-Otsu	**0.773**	**0.803**	**0.788**	4.123
TCN-3σ	0.472	**0.803**	0.594	**4.120**
LSTM-Otsu	0.522	0.576	0.547	7.463
LSTM-3σ	0.338	0.409	0.37	7.460

V. EXPERIMENTS

A. Study Area and Experimental Data

The study area, Loess Plateau, is located at 33°41′N to 41°16′N and 100°51′E to 114°32′E, spanning parts of Ningxia Hui Autonomous Region, Shanxi, Inner Mongolia, Shaanxi, Gansu, Qinghai, and Henan provinces (Fig. 5). It stretches more than 1000 km from east to west and 750 km from north to south and covers an area of approximately 640 000 km^2. The Loess Plateau is in a transitional zone from plain to plateau and forest to grassland (Fig. 6). The vegetation types include warm-temperate deciduous broad-leaved forests, subtropical evergreen broad-leaved forests, alpine vegetation on the Qinghai–Tibet Plateau, and temperate deserts and grasslands (Fig. 7). Its natural environment is complex, it is a key area of soil and water conservation in China, and it is a hot topic of current research on land-cover changes [51].

The experimental data are 16-day-synthesized MOD13Q1-EVI data collected from 2001 to 2020, with a spatial resolution of 250 m, including 460 time phases, each of which contains four frames of h25v05, h26v04, h26v05, and h27v05 (Fig. 8). After mosaicing and cropping according to the boundary of the Loess Plateau, the total data volume is approximately 34.7 GB, which can be called a typical large-scale and long-term remote sensing dataset. In addition, the necessary cloud cover removal and missing value filling are conducted by referring to previous research [33].

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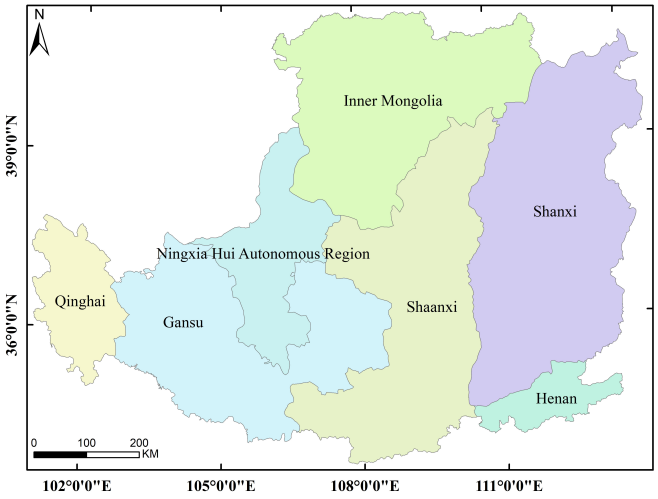

Fig. 5. Provincial administrative division map of the Loess Plateau.

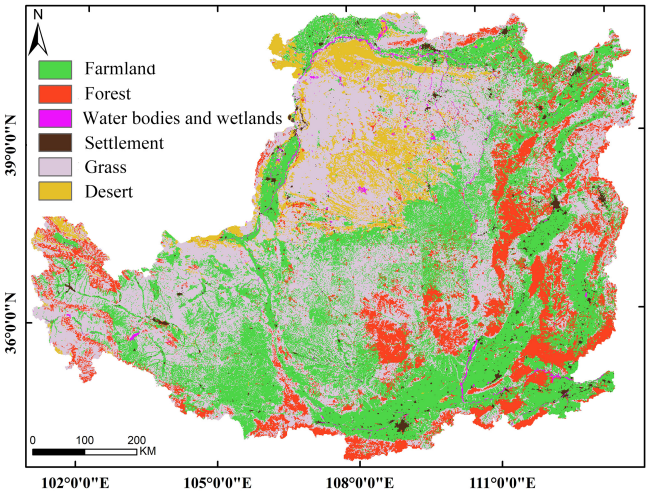

Fig. 6. Land-cover map of the Loess Plateau in 2010 (1:100, 000) [52].

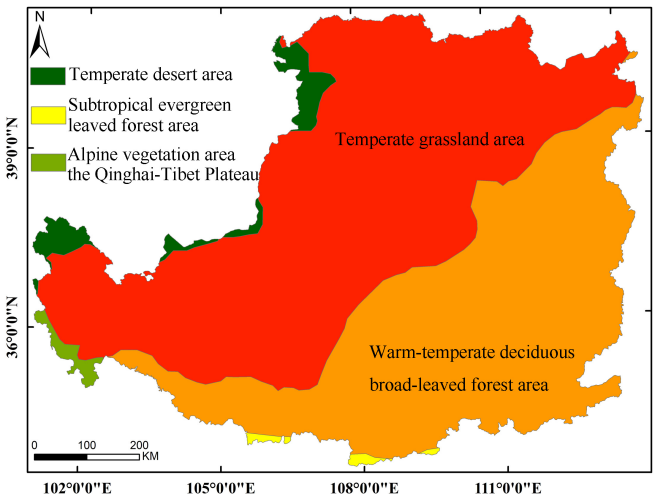

Fig. 7. Vegetation regionalization map of the Loess Plateau in 1989 (1:100, 000) [53].

B. TCN-Based Remote Sensing Time-Series Prediction

In TCN-based remote sensing time-series prediction, that is, taking the 23 time-phase MOD13Q1-EVI data in the first year as the basement, we use causal convolution to iteratively extract the time-series relationship of historical data, thereby sequentially predicting the subsequent MOD13Q1-EVI value. Given that this study uses a pixel-based time-series prediction model, the TCN model of each pixel needs to be trained and predicted separately. Therefore, to improve the processing efficiency of the experimental data, this study divides the entire Loess Plateau image into 15 tiles, each of which contains approximately 1000 × 1500 pixels on average. Then, 15 RTX 2080 Ti GPU cards are used to process these small tiles in parallel, and the iterative training times setting of the TCN model in each small tile still adopts the early stopping strategy to prevent model overfitting.

The TCN-predicted results are highly consistent with the actual value, which can be proved by the RMSE, PCC, and SDTW values listed in Table II, that is, for the time-series prediction of a single pixel, the TCN model can meet the high-precision fitting requirement.

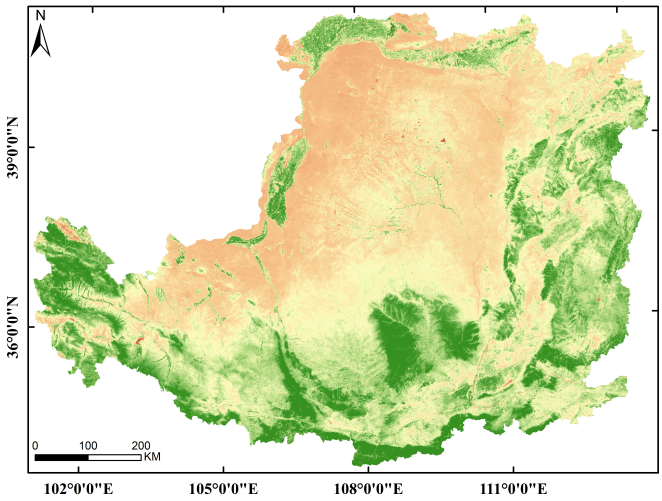

Fig. 8. Pseudo-color image of MOD13Q1-EVI of the Loess Plateau on June 26, 2001.

C. Similarity-Measurement-Based DL Model Migration

It is very time-consuming to use the TCN model for data fitting and forecasting pixel-by-pixel, which is not conducive to large-scale data processing. Therefore, we propose to adopt the similarity-measurement-based DL model migration method. In this experiment, a single small block of 1000 × 1500 pixels can be automatically clustered into 20 categories according to the SDTW distance as the similarity measurement criteria; thus, each category only needs to randomly choose any one timing curve for TCN model training and then migrate the trained model to other time-series curves of the same category for prediction, for the time series within the category which has a relatively high similarity. To ensure the prediction accuracy of the migration model, we still need to use the SDTW distance to measure the similarity between the predicted and actual values of the migration model. In this experiment, we set the lower limit of the similarity threshold to 10. If the SDTW distance between the predicted and actual values exceeds the preset threshold, the time-series curve needs

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TABLE II

ACCURACY EVALUATION OF THE TCN-PREDICTED AND ACTUAL VALUES. PIXEL COORDINATES REPRESENT THE SPATIAL POSITION OF THE PIXEL; THE ORIGIN OF THE COORDINATES IS AT THE UPPER LEFT CORNER OF THE MINIMUM BOUNDING RECTANGLE OF THE REMOTE SENSING IMAGE, THE POSITIVE x-AXIS EXTENDS TO THE RIGHT, AND THE POSITIVE Y-AXIS EXTENDS DOWNWARD

Tile number	Pixel coordinates	RMSE	PCC	SDTW
1	530,2600	0.0049	0.9996	3.3801
2	3715,415	0.0136	0.9999	5.6569
3	5065,608	0.0114	0.9999	5.2915
4	160,1760	0.0176	0.9998	4.0035
5	2260,1800	0.0142	0.9999	7.8286
6	3778,1260	0.0132	0.9998	7.6758
7	4948,1330	0.0154	0.9998	3.5332
8	400,2600	0.0179	0.9998	5.7763
9	2240,2770	0.0154	0.9998	6.9282
10	3630,2400	0.0172	0.9998	7.3923
11	4821,2343	0.0128	0.9998	6.3281
12	520,3030	0.0195	0.9998	7.4833
13	2270,3370	0.0274	0.9998	5.8044
14	3647,3230	0.0282	0.9998	4.899
15	4760,3280	0.0236	0.9998	4.899

to be classified into a new category, the migration model is discarded, and a new DL model training is performed again. Then, the model migration prediction and accuracy evaluation will be performed again. If the similarity is lower than our preset threshold, then the classification is repeated, and this procedure is followed until all pixels have reached the prediction accuracy. In addition, to ensure that the algorithm can converge, we set the maximum number of migration classification groups as 200. If the total number of categories reaches the maximum value, but the accuracy of individual timing prediction still does not reach the preset threshold, the algorithm will automatically end. It should be noted that the initial clustering parameter 20 was determined according to the spatial resolution of the remote sensing images and the land-cover types in the study area, and it was recommended to be set to two to three times the number of land-cover classes; in addition, after repeated experiments, we found that if the initial clustering parameter was too small, the prediction result of the transferred DL model would be inaccurate, resulting in a large number of subcategory resplitting operations. If it was too large, the clustering convergence speed would be greatly reduced; the SDTW-based similarity threshold 10 is an empirical parameter, which needs to be determined according to the characteristics of actual data; and the maximum number of migration classification groups 200 is also determined according to the spatial resolution of the remote sensing images and the land-cover types in the study area, which can be set to an integer multiple of the initial clustering parameter. Finally, we found that with the help of Python multiprocessing [54] and DL model early stop strategy, it took about 57.3 h for all time series to complete fitting and prediction on each small tile using our proposed migration prediction model. In addition, to quantify the overall prediction accuracy, the RMSE, PCC, PSNR, and SSIM results between the actual and TCN-predicted values are calculated (Fig. 9).

Fig. 9 shows that, except for a few small areas, the RMSE values in most areas are less than 1.0 or close to 0, the PCC values are larger than 0.8 or close to 1, the PSNR values at all time points are greater than 65 dBs, and the SSIM values are larger than 0.998 or close to 1, which strongly illustrate that the obtained results are slightly different from the actual values, both in spatial and time dimensions. In other words, the proposed transferred DL model can well realize the high-precision fitting and prediction problem of large-scale time-series data.

D. Otsu-Based Adaptive Time-Series Change Detection

After using the transferred DL model to obtain the time-series prediction value, the adaptive threshold can be used to evaluate the difference between the prediction and the observation time series to detect the change points. As each time series is individually calculated based on its own data distribution characteristics, the adaptive threshold value will not cause interference between each time series. The final annual change points of ground cover on the Loess Plateau from 2002 to 2020 are shown in Fig. 10. Note that as we set 2001 as the base year, there are no change points, and if the same geographic location changes multiple times each year, only the first time change will be displayed.

Fig. 10 shows that, in the spatial domain, there are change points in almost every corner of the Loess Plateau; in the time dimension, the change points are mainly concentrated during June–August (Fig. 11), which might be because these three months are the maturity or harvest time of crops in the Loess Plateau.

However, due to the wide area of the Loess Plateau, the crop maturity and harvesting time also vary considerably. In addition, the climate and vegetation types in the Loess Plateau are diverse, and the resulting land covers are much different. Therefore, a simple macroanalysis cannot determine the characteristics of land-cover change and clarify the drivers; thus, detailed time and space dimension analysis is still required to learn more lessons.

E. Lessons Learned From the Detected Change Points

1) Spatial Distribution Characteristics of Change Points: The spatial distribution feature analysis takes the natural month as the unit, counts the frequency of change points pixel-by-pixel, and accumulates all years to explore the spatial characteristics of frequent change points in each month. The final result of the spatial distribution of the detected change points every month is shown in Fig. 12.

The following lessons can be learned from Fig. 12.

1) On the whole, the spatial distribution characteristics of change points have a strong correlation with the land-use type and the boundary of vegetation regionalization.
2) In March, the change points are concentrated in a small part of the southern part of the warm-temperate deciduous broad-leaved forest area, close to the highway and the county seat; in April as well, the change points are

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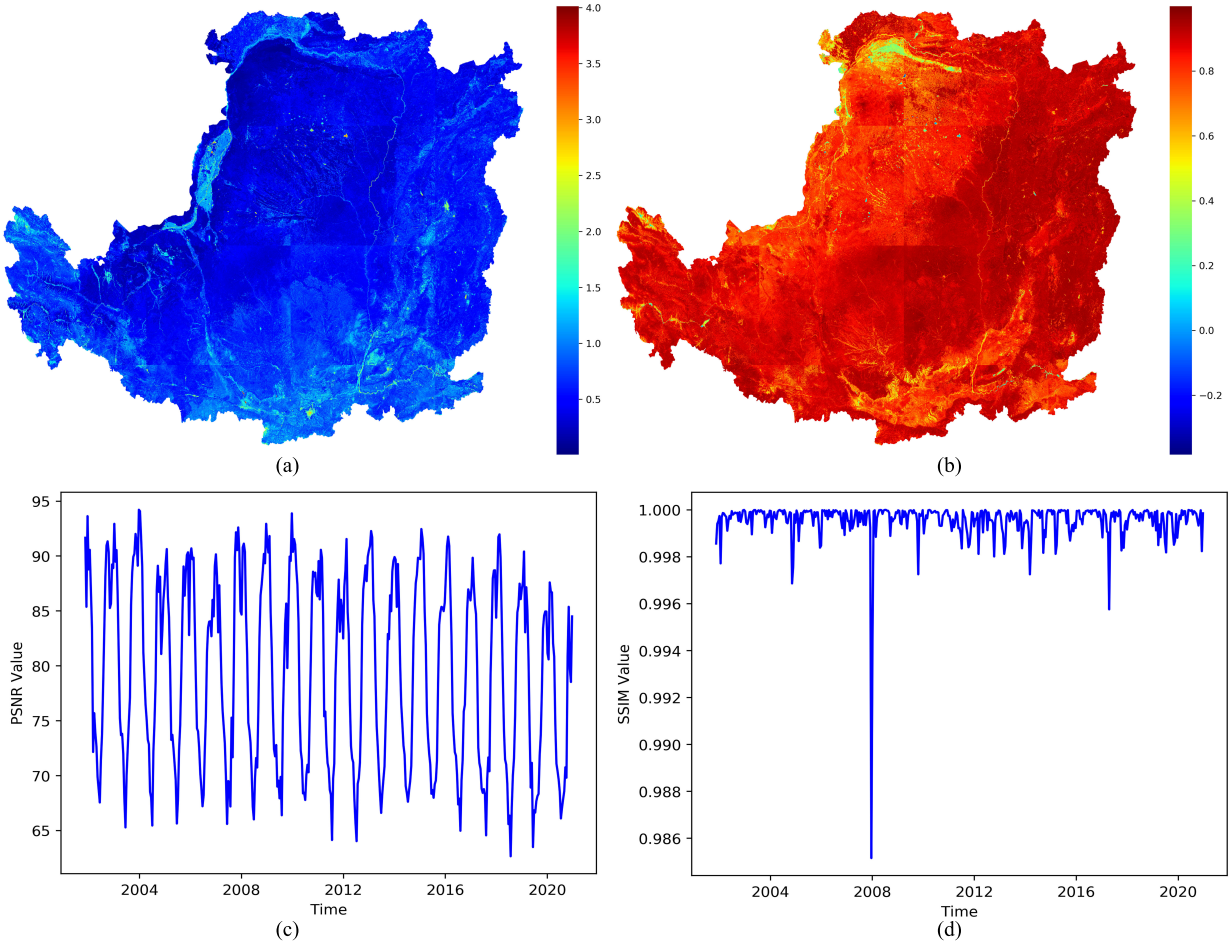

Fig. 9. (a) RMSE, (b) PCC, (c) PSNR, and (d) SSIM results between the actual and TCN-predicted values.

distributed in the southern part of the warm-temperate deciduous broad-leaved forest area, near the county seat and the highway. By comparing the land-cover map in 2010, it can be concluded that the land-use types of the areas where the changes occur in March and April are farmland and deciduous broad-leaved forest areas, double-cropping planting area, and humid and semihumid zones. Therefore, these changes mainly result from the following.

a) The rapid increase in the vegetation index curve amplitude caused by the creeping growth to the rapid growth stage of winter wheat after turning green.
b) The vegetation index changes caused by the deciduous broad-leaved forest germination and growth of new leaves.
c) Land-cover changes caused by human activities, such as spring cultivation and tree plantation.

3) In May, the change points spread to the entire warm-temperate deciduous broad-leaved forest area and the southwest and northwest of the temperate grassland area. They are still mainly distributed in agricultural land and forest areas along the highway. The following are the possible reasons for this.

a) In the farmland of the temperate deciduous broad-leaved forest area, the winter wheat matures and gradually turns to withered, which shows a rapid decline in the vegetation index curve.
b) For forests in temperate deciduous broad-leaved forest regions, the leaf area gradually increases and the overall canopy closure increases, which is reflected in a substantial increase in the vegetation index curve.
c) In the northwestern part of the temperate grassland region, which is the irrigation area for diversion from the Yellow River in Ningxia, large areas of crop plantation and rapid growth have caused changes in the land cover.
d) In the southwestern part of the temperate grassland region, which is the farming area of Qinghai, the plantation and rapid growth of crops causes changes in the land cover.

4) In the three months of June–August, with the exception of some desert areas, the entire Loess Plateau has undergone significant changes in land cover. In terms of frequency, an average of ten changes have occurred in 20 years. This is mainly because of the following.

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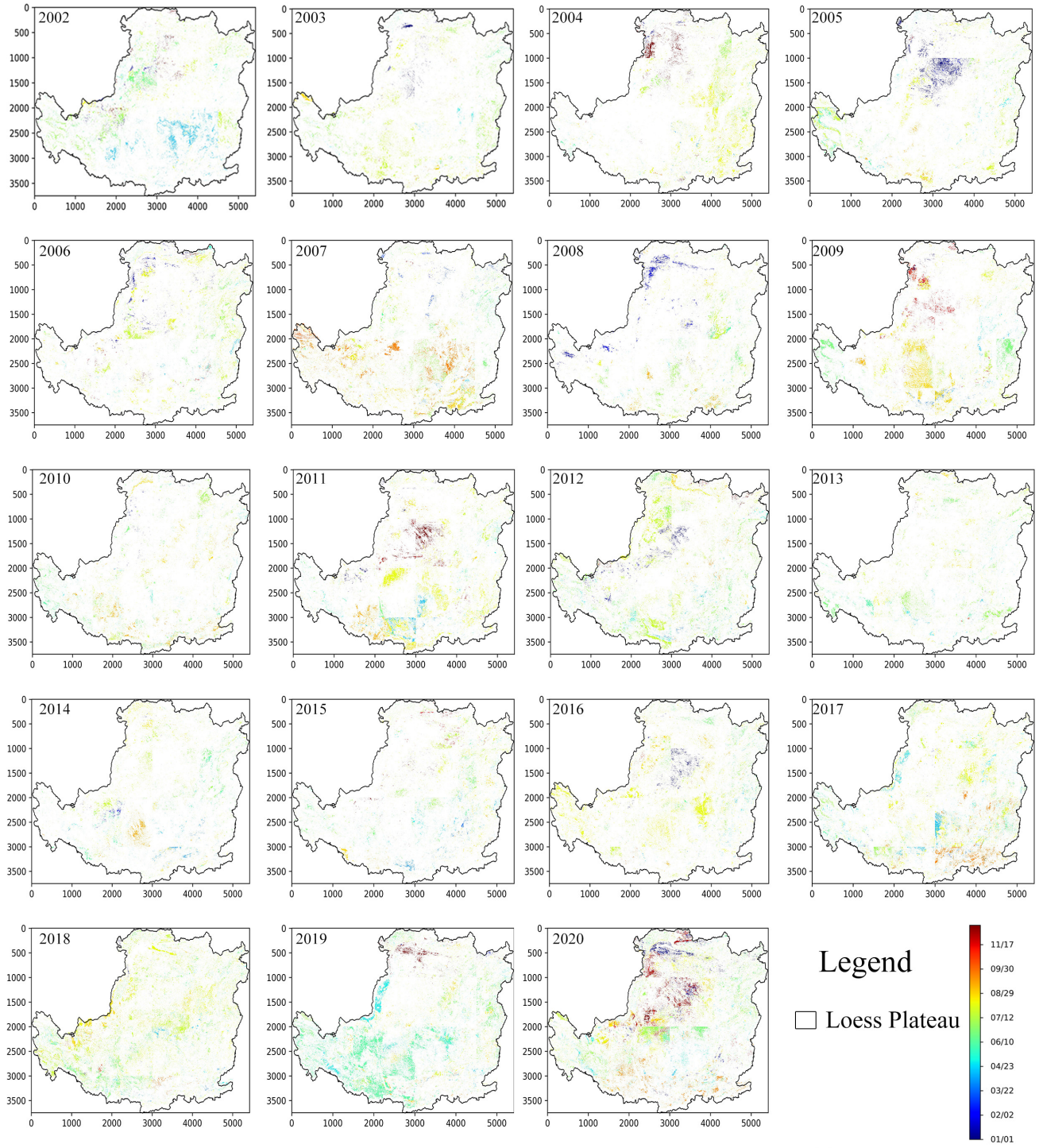

Fig. 10. Annual distribution map of change points detected in the Loess Plateau.

a) The southern region of the warm-temperate deciduous broad-leaved forest is a double-cropping planting area, where large-scale crop harvesting has caused changes in the land cover.
b) The forests and crops in the entire Loess Plateau reach the growth peak during this period, and the vegetation curves show a significant increase compared to the previous months.
c) The harvesting of pastures in the pastoral areas of Inner Mongolia has caused changes in the land cover. In addition, in July and August, there are low-frequency change points in the desert coverage area. This is mainly due to the changes in the vegetation index curve caused by the artificial forest farms planted under the Three North Shelterbelt Project led by the Chinese government.

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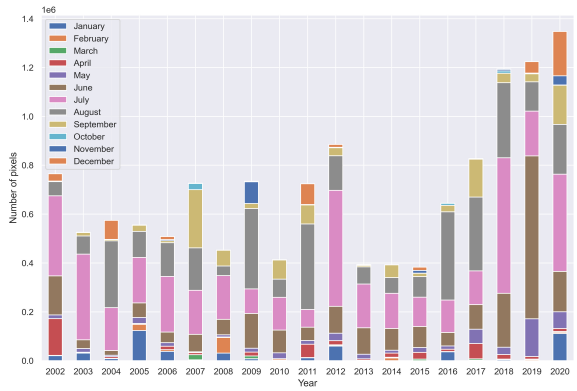

Fig. 11. Annual statistical map of change points detected in the Loess Plateau region by month.

5) In September, the change points in the central and northern parts of the temperate grassland area decreased significantly, and there were still many change points in parts of the southwest and northwest. The change points in the temperate deciduous broad-leaved forest area were sparse, and the average change frequency in the 20-year period decreased. This is mainly because of the following.
   a) The central and northern parts of the temperate grassland area are deserted and landed with sparse vegetation, and the grassland was degraded in September.
   b) The southwest and northwest of the temperate grassland region are farmland and forest distribution areas, and crop harvesting is an important factor causing land-cover changes.
   c) The southern part of the temperate deciduous broad-leaved forest area is double-cropping planting area, and crop harvesting is the main factor that causes land-cover changes in this area.
   d) In the central and northwestern regions of temperate deciduous broad-leaved forests, land-cover changes might be caused by factors such as crop harvesting and forest deciduousness.
6) In October, change points were mainly concentrated in the forest coverage area of northern Qinghai Province in the temperate desert area, possibly due to the land-cover changes caused by snow accumulation and snowmelt. The northern part of Ningxia is the Yellow River irrigation area, which is caused by rice harvesting in late September. In southern Shaanxi, southwestern Shanxi, and parts of Henan, all change points are located near the county resident, which indicates land-cover changes caused by autumn crop harvesting, land plowing, and winter wheat sowing.
7) In November–January, change points were mainly concentrated in the grassland- and desert-covered areas in the temperate grassland area, which are far away from the city. This might be caused by long-term ground snow or snowmelt after the grassland has withered.

2) Time Distribution Characteristics of Change Points: The analysis of time distribution characteristics takes province as a unit and counts change points of each province monthly to explore the occurrence time of the detected change points in various provinces. The annual statistical map of change points detected in each province by month is shown in Fig. 13.

The following lessons can be learned from Fig. 13.

1) The months with the maximum occurrences of change points in all provinces are 6–9 months. This is mainly because of the generally high temperature and maximum rainfall in the Loess Plateau during these four months [55]. The vegetation reached its growth peak and gradually began to wither, the crops began to be harvested, and the leaves gradually changed from dark green to golden, which caused the vegetation index curve to change from high to low.
2) In Henan, Shaanxi, and Shanxi, in addition to the significant changes in the four months of June–September, there were also significant land-cover changes in April. The following are the possible reasons for the changes in these three provinces during April-September.
   a) Henan and parts of Shaanxi and Shanxi are located in the Loess Plateau, double cropping is performed every year, and in April and May, the vegetation index curve changes because of the winter wheat turning green and rapidly growing.
   b) In June–September, the vegetation index curve changes because the crops or vegetation reach their growth peak or harvesting.
3) Qinghai has the most frequent change points in June and July, which may be related to the peak growth of crops and forests in these months.
4) Changes in Ningxia and Gansu are mainly concentrated in July and August. The growth of crops or vegetation reaches its peak, rice and other crops change from dark green to golden, and crops are harvested.
5) The change points in summer are attributable to factors such as the growth peak and harvesting of the grassland; in winter, and they are attributable to long-term snow accumulation and melting snow.

To further sort out the distribution law of each change point in the time dimension, this article proposes to use the pixel-by-pixel time-series autocorrelation coefficient [56] to calculate the periodicity of each time-series change point. The autocorrelation function (ACF) is used to measure the correlation between the observations of every k time units (y_t and y_{t-k}) in the time series. If the correlation is strong, then k can be considered as the period of the time-series change point.

The ACF calculation can be performed as follows:

$$\mathrm{ACF(k)} = \rho_k = \frac{\mathrm{Cov}(y_t, y_{t-k})}{\mathrm{Var}(y_t)} = \frac{E((y_t-\mu)(y_{t-k}-\mu))}{E(y_t-\mu)^2} = \frac{\sum_{t=k+1}^{N}(y_t-\mu)(y_{t-k}-\mu)}{\sum_{t=1}^{N}(y_t-\mu)^2} \tag{7}$$

where y_t is the time series, ρ_k is the autocorrelation coefficient at k lag time unit, k is the lag time unit, Cov is the

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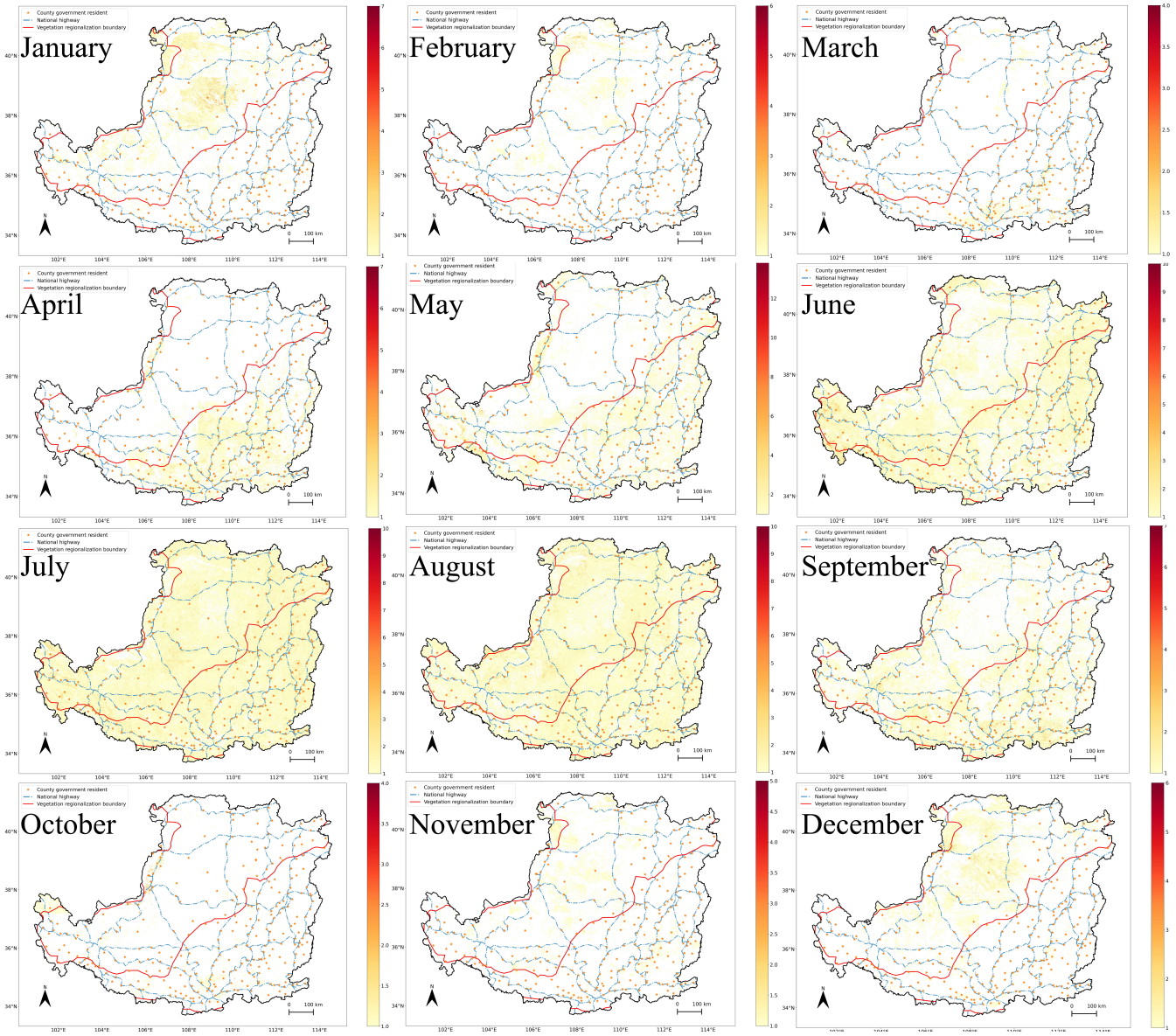

Fig. 12. Spatial distribution of detected change points every month from 2002 to 2020. The color bar on the right of each image represents the frequency of change points. The darker the color, the more changes are considered to have occurred between 2002 and 2020.

autocovariance of the time series, Var is the variance of the time series, and μ is the mean of the time series.

The pixel-by-pixel calculation of the ACF value of change points in the Loess Plateau area from 2002 to 2020 is shown in Fig. 14.

Fig. 14 indicates the following.

1) A large part of change points have a time correlation of 23, that is, the cycle takes a year, and the main distribution area is the cultivated land area of the Loess Plateau. This is consistent with the annual cropping rule for most crops in the entire Loess Plateau.
2) The change points with a half-year or eight-month cycle are scattered throughout the Loess Plateau, mainly concentrated in county residences and highways in the semihumid area of the Loess Plateau. This is attributed to human activities, such as sowing and harvesting, in the cultivated area.
3) The multiyear periodic change points are mainly distributed in Qinghai, northeastern Gansu, central Inner Mongolia, southwest Shanxi, and parts of Henan. These areas are mainly covered by forests or deserts, which is attributable to natural phenomena, such as forest deciduous and snow accumulation.

3) Discussion: Through the above analysis of the temporal and spatial distribution characteristics of the change points in the Loess Plateau detected from 2002 to 2020, the following lessons can be learned.

1) The time distribution of the change points in the entire Loess Plateau is concentrated in the summer with abundant rainfall and high temperature (Fig. 12). This is mainly attributed to factors such as the crop growth period reaching its peak and beginning to decline and crop harvesting. From the perspective of spatial dimension, the area where the change point occurs is closer

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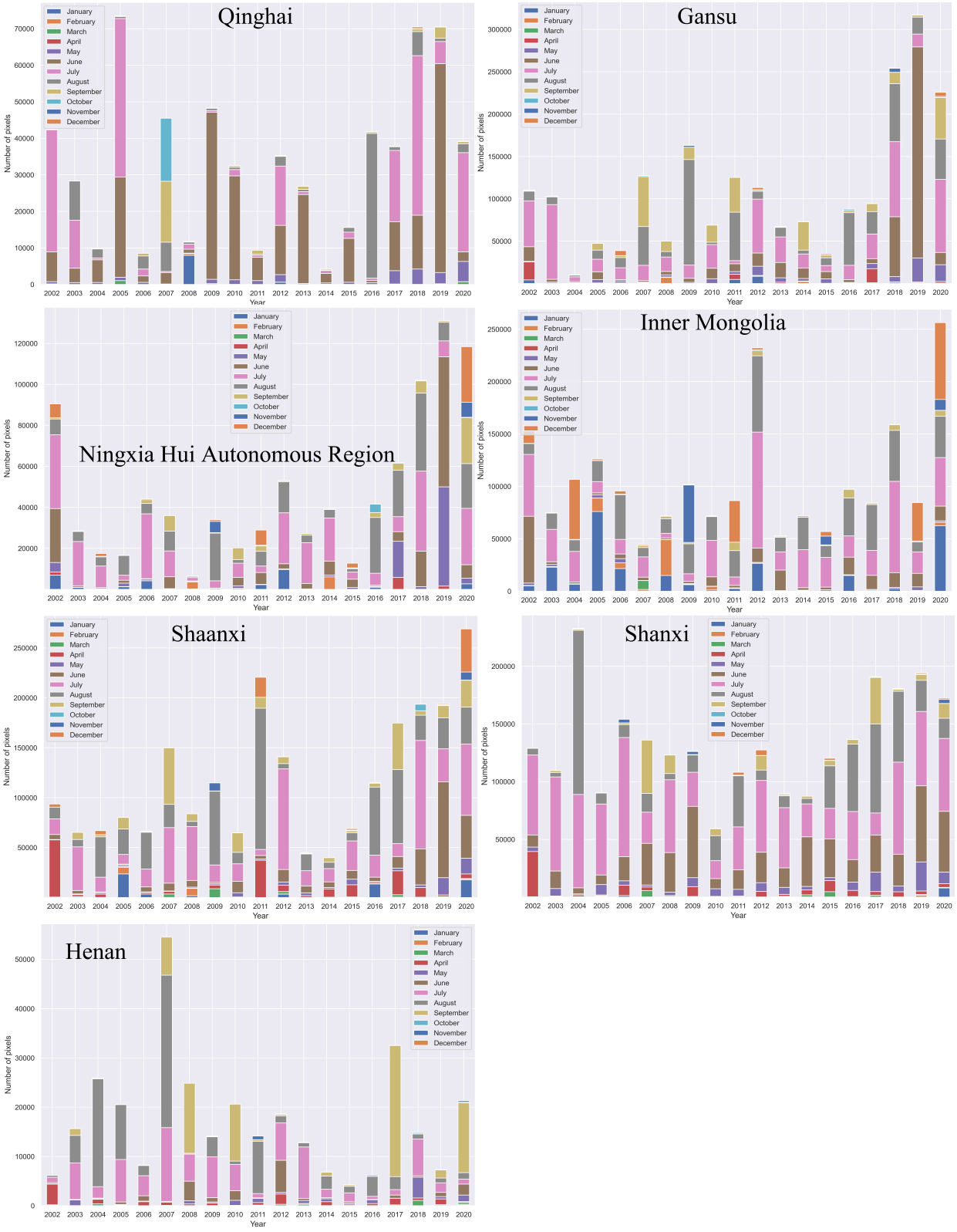

Fig. 13. Annual statistical map of change points detected in each province by month.

to the county residence and highway (Fig. 12) and is a cultivated area with densely populated population and frequent human activities. Therefore, we conclude that the main drivers of land-cover changes in the Loess Plateau are the human activities under natural conditions (such as temperature and precipitation).

2) From the perspective of the entire timeline (see Fig. 11), from 2002 to 2015, except for 2007, 2009, 2011,

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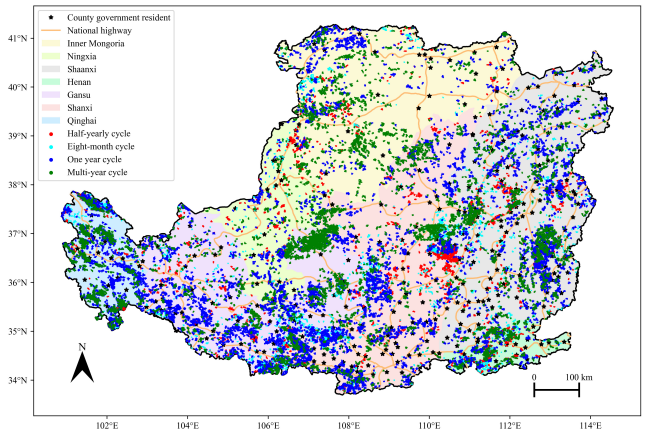

Fig. 14. Spatial density cluster map of the half-yearly, eight-month, one-year, and multiyear-periodic change points.

and 2012, the number of change points in the Loess Plateau has stabilized as a whole. This is mainly due to the artificial forest farms and pastures produced by the Chinese government-led project of returning farmland to forests and grasslands since 1999 have gradually matured from seedlings after more than ten years of development. The large-scale changes in 2007, 2009, 2011, and 2012 mainly occurred in the central and northern part of Shaanxi Province and its surrounding areas (see Fig. 10), which may be caused by the special planning projects for consolidating the results of returning farmland to forests and preventing wind and sand fixation.

3) From 2016 to 2020, the number of change points in the entire Loess Plateau, especially in Qinghai, Gansu, Ningxia, and Shaanxi, increased yearly (Fig. 13). This is mainly because the ecological restoration projects of land covers, forests, fields, lakes, and grasses led by the Chinese government since 2016 have played a huge role in protecting forests and grasslands [57].
4) According to the change point time correlation results (Fig. 14), although most of the change points show a time distribution cycle consistent with the crop planting rule, there are still change points that should be a one-year or half-year cycle and are divided into multiyear cycles. This is mainly because the data selected in this experiment are the MODIS-EVI data synthesized in 16 days. The data synthesis process smoothens out part of the change information accumulated within the 16 days, so the change points that have occurred in many years might not be detected.

VI. Conclusion

In order to realize high-precision monitoring of large-area land-cover time-series changes, this study proposed SDTL-TSACD model. Specifically, in order to solve the problem of low change detection accuracy of linear fitting and prediction, the TCN-based remote sensing time-series prediction method is proposed, and an early stop strategy is used to prevent overfitting. Aiming at the problem that the DL model takes a long time to process large-scale data, the similarity-measurement-based DL model migration strategy is proposed, that is, the SDTW distance with strong noise resistance is proposed to measure the time-series similarity, which is not only used to divide the transferable range of the DL model but also used to evaluate the prediction accuracy of the transferred DL model. Considering the difficulty in determining the threshold of unsupervised time-series change detection, the Otsu adaptive threshold is adopted to determine time-series change points, and the spatial neighbor relationship is also used to eliminate pseudo-change points. Finally, 132 benchmark datasets are used to validate the performance of change point detection, and the large-scale time-series remote sensing data of the entire Loess Plateau in past 20 years are used to verify the actual application. The results show that the proposed SDTL-TSACD model can effectively identify a series of information on land-cover changes in the Loess Plateau resulting from natural phenomena and human activities in the past 20 years and clarify that the driver is human activities under natural conditions.

However, because the experimental dataset MODIS-EVI is a 16-day synthesis data, the data synthesis process smoothens considerable change information. The case study on the Loess Plateau does not detect deforestation and fires in a small area, as well as the lost information on faint changes on a smaller time scale. Therefore, the use of original observation data with higher temporal and spatial resolutions for monitoring long-term remote sensing of large areas and to further improve the calculation efficiency and generalization capabilities of the SDTL-TSACD model still remains the focus of future research.

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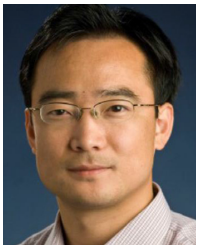

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